

RoomBridge Supporting Mental Health and Wellbeing through Accountable AI Mediation in PolyU Halls

Keywords: AI mediation; student mental health; wellbeing; roommate relationships; needs preservation

a. Background/Introduction

Housemate relationships can provide support or contribute to distress (Foulkes et al., 2021). Poor accommodation relationships and low belonging are associated with poorer mental health (Worsley et al., 2023), while supportive social networks feature in a systematic review of student wellbeing (Campbell et al., 2022). These findings motivate examining hall relationships as one setting for care.

LLMediator and subsequent dispute-resolution evaluations explore AI-assisted mediation (Westermann et al., 2023; Tan et al., 2024). NegotiationArena examines bargaining between LLM agents (Bianchi et al., 2024), while medical-dispute counselling and consensus frameworks extend this research (Kim et al., 2025; Dong & Ito, 2026). RoomBridge addresses a specific question within this emerging field: whether apparently considerate roommate agreements silently omit residents' stated needs.

b. Objectives

The study will test whether recording, auditing and revising around stated needs reduces omissions and unequal coverage between residents. A secondary objective explores whether residents feel heard and experience fairer negotiations. The hypothesised wellbeing pathway links clearer agreements with less conflict-related stress and sleep disruption. Need coverage and mental health outcomes will be evaluated separately.

An agreement about visitors, for example, should explicitly address a roommate's sleep requirement and identify unresolved trade-offs. The proposed contribution is an auditable link between each person's expressed needs and the final wording. Residents' feedback would distinguish textual acknowledgement from arrangements they consider workable and respectful in everyday hall life.

c. Methods

Four conditions will be compared across 22 authored synthetic scenarios: generic mediation, context-informed mediation, multi-agent deliberation and RoomBridge. Each scenario contains a fixed set of stated needs. Repeated runs will record prompts, models and seeds. Comparisons will be paired within scenarios, with uncertainty estimated across scenarios; an additional matched-information comparison will examine differences in access to hall rules.

RoomBridge records needs individually and generates candidate agreements. A separate auditor checks each need against agreement text without seeing the mediator's reasoning. Positive judgements require supporting quotations. Missing, contradicted or unresolved needs trigger bounded revision. Policy and assumption checks identify prohibited arrangements and invented preferences. Threats and serious safety concerns should halt mediation for human support.

Measures will include graded need coverage, omitted or contradicted needs, and coverage differences between residents. Human reviewers will independently assess a held-out sample. An exploratory resident pilot, subject to ethics approval and consent, would compare perceived fairness, stress, sleep disruption and wellbeing before mediation and at follow-up. A usual-support comparison group and interviews would help interpret changes and assess voluntary adoption. Validated measures would be used where appropriate, personal information minimised, and residents could reject suggestions.

d. Expected Results/Findings

Offline tests demonstrate recovery of deliberately omitted needs and escalation in designed cases. Live-model evaluation will test whether these mechanisms reduce omissions consistently. The proposed pilot would investigate possible wellbeing benefits; clinical effectiveness has not been established. Disagreements that remain unresolved will be reported explicitly, since mentioning every need does not guarantee an acceptable settlement.

e. Conclusion/Discussion, Implications, and Future Directions

Aligned with Sub-theme 3, Social Science, Healthcare Innovations, RoomBridge connects care for life in the AI era with residents' agency and supportive hall relationships. For PolyU's development in 2026, its contribution is a testable process that helps hall staff identify overlooked concerns while keeping decisions with residents. Counselling and clinical care remain responsibilities of qualified professionals.

References

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